

BRUK ABU

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Experience

- Venture MMO** | *Founder, Lead Developer* **Summer 2020 - Present**
- Spearhead Roblox MMO development with 100K+ lines of code & a closed alpha with 100+ testers, collaborating with major YouTubers whose videos received 100K+ views and millions of subscribers.
 - Design and implement key end-to-end systems including global economy, AI, matchmaking, and player progression through database, ensuring smooth gameplay across hundreds of concurrent players.
 - Exercised object-oriented programming practices such as inheritance to create modular and scalable systems, reducing development time for new features by 30%.
 - Co-lead a cross-functional team of 7 developers, coordinating milestones and agile sprints to deliver new features and updates ahead of schedule.
 - Foster a vibrant community of 10,000+ Discord members by sharing development progress, sneak peeks, and actively engaging with player feedback to guide feature priorities.
- Northeastern University** | *Teaching Assistant* **Sep. 2023 - Dec 2023**
- Fundamentals II: Introduction to Class-based Program Design*
- Supported and mentored over 500 students in CS2510, Northeastern's largest core Java programming course, emphasizing object-oriented programming principles and advanced language features.
 - Guide students through challenging coding problems in office hours, encouraging critical thinking and problem-solving skills.
 - Hosted weekly lab sessions attended by dozens of students; consistently received positive feedback regarding clarity and usefulness of support.
 - Facilitate learning and understanding of complex concepts such as abstract classes, interfaces, and other advanced Java programming techniques.
- United Health Care: Optum AI** | *Full Time Co-Op* **July 2022 - December 2022**
- Led data visualization, data science, and data analytics efforts to communicate model results and findings to team members and stakeholders.
 - Applied natural language processing techniques to extract insights from electronic medical records.
 - Developed and implemented machine learning models for predicting patient outcomes and disease progression.
 - Utilized TypeScript and Angular to develop the front-end interface for multiple machine learning applications, resulting in a more robust and maintainable codebase.

Projects

- Resell Bot** | *Software Developer* **January 2021 - March 2021**
- Developed and implemented an automated purchasing system using Python and Selenium, reducing manual operation time by 90% and increasing success rates by 20%.
 - Engineered a multi-profile integration system that increased item acquisition success rates by over 150% through parallel processing.
 - Implemented RESTful API integration using JavaScript for real-time data acquisition and automated purchase requests.
- Bluemont Bikes** | *Head of Technologies, Full Stack Developer* **October 2020 - March 2021**
- Architected and developed a full-stack web application for bike repair services, implementing responsive design principles and modern UI/UX practices.
 - Engineered backend systems using PHP and MySQL to handle service requests, automated email notifications, and real-time user updates.
 - Optimized user engagement metrics by 40% through implementation of intuitive user interfaces and streamlined customer service workflows.

Technical Skills

Languages: Lua, Python, Java, HTML/CSS, JavaScript, TypeScript, SQL, C++, PHP
Developer Tools: VS Code, Eclipse, Roblox Studio, Jenkins, Docker, Azure, Postgres, MongoDB (NoSQL), AWS, Apache
Technologies/Frameworks: React, Node, Jest (QA Testing), Next.js, Kubernetes, Next.js, JSON, REST APIs, Linux, GitHub, JUnit, WordPress, CI/CD

Relevant Coursework

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|----------------------------------|--------------------------|-------------------------|-------------------------|
| • Algorithms and Data Structures | • Object-Oriented Design | • Logic and Computation | • Computer Architecture |
| | • Discrete Structures | • Systems Programming | • Networking |
| | • Computer Systems | • Web Development | |

Education

Northeastern University **Sep. 2020 – June 2024**
Khoury College of Computer Sciences
Bachelor of Science in Computer Science with concentration in Software Engineering
Honors: 3.3 GPA

Languages

Spanish - Fluent, Amharic - Fluent, French - Conversational, Japanese - Elementary